Các công nghệ lập trình hiện đại

Docker và WSL2

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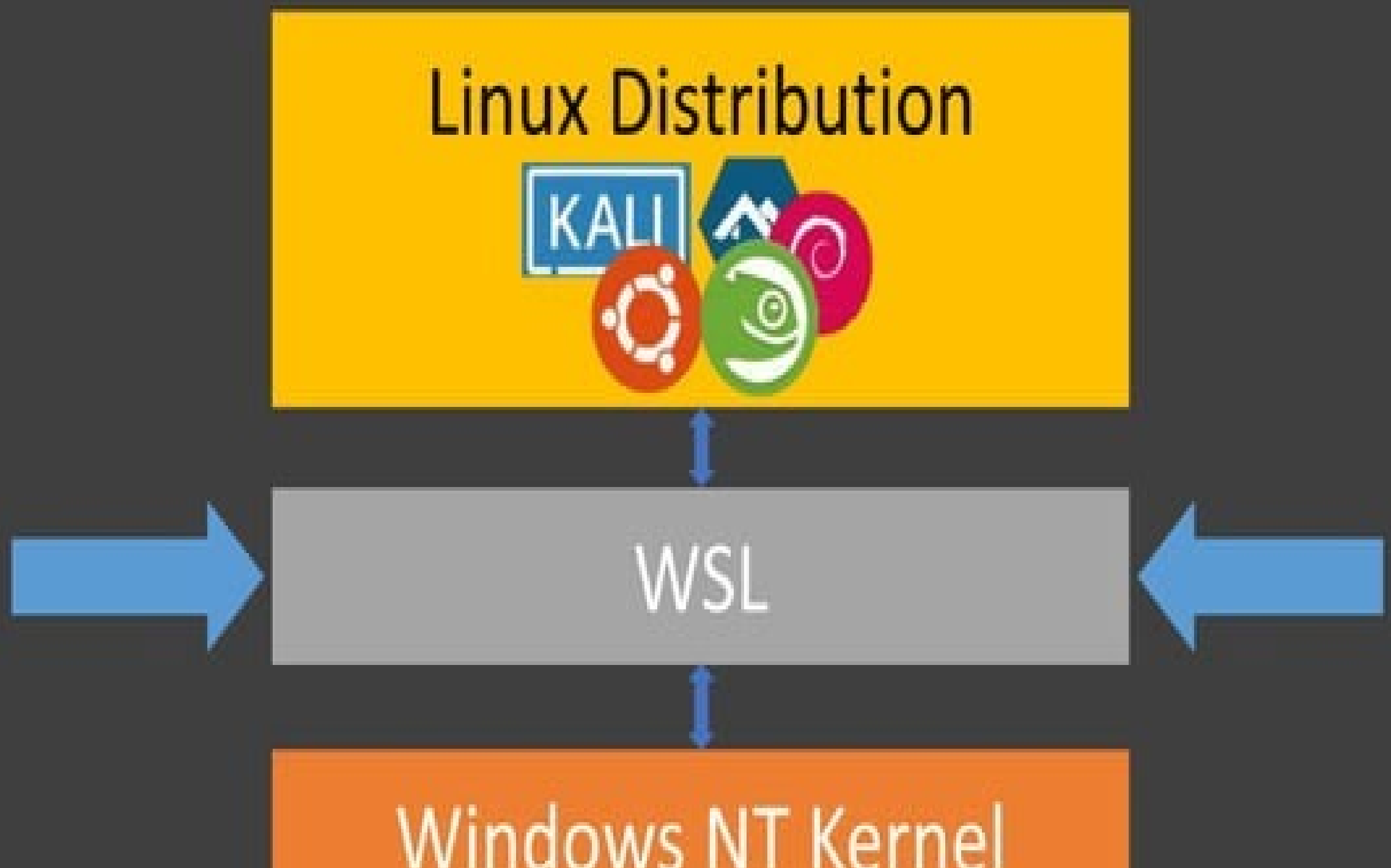
WSL 2: Windows Subsystem for Linux v2

How does this work? – WSL 1 Architecture



Renaming a directory with an open file

Windows

OpenFile("C:\\oldname\\file.txt", ...);

MoveFile("C:\\oldname",
"C:\\newname");

ERROR_ACCESS_DENIED

Linux

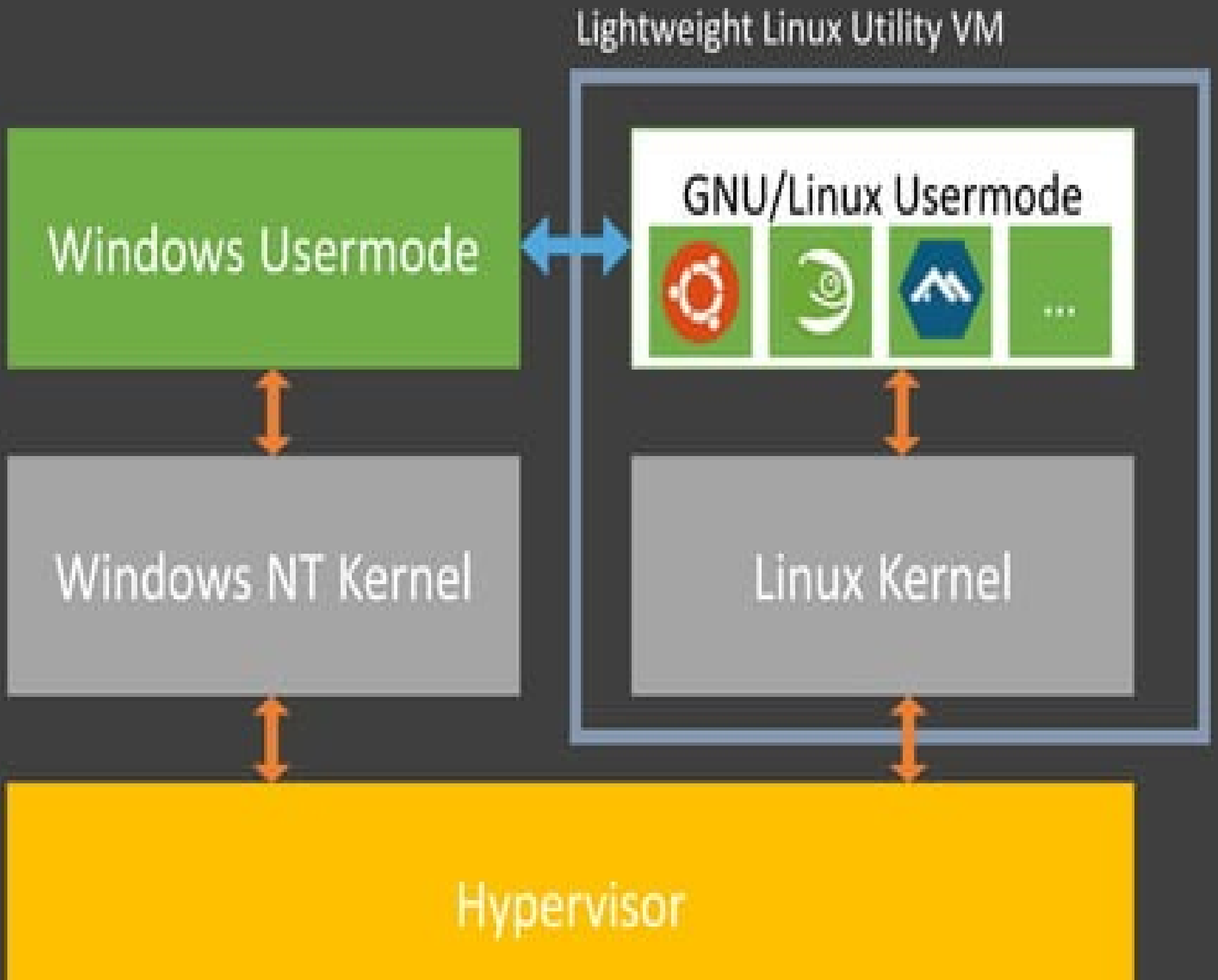
open("/oldname/file.txt", ...);

rename("/oldname", "/newname");

SUCCESS


https://www.flickr.com/photos/jev55/8093515962/

WSL 2 Architecture Overview



Traditional VM

- Isolated
- Slower to boot
- Large memory footprint
- You need to manage it

vs

WSL 2

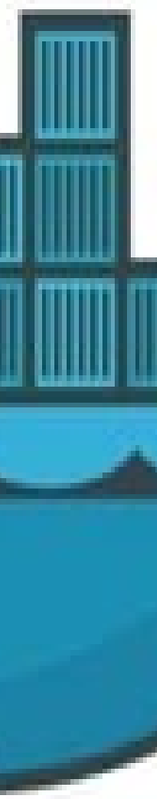
- Integrated
- Fast to boot (~1 second)
- Small memory footprint
- Only runs when you need it

File System Performance

WSL 2 speed comparison test WSL 1 on a Surface Laptop

git clone	npm install	cmake
2.5x	**4.7x**	**3.1x**
faster	faster	faster

Full System Call Compatibility



docker engineering

Why Docker? Products Solutions Customers Resources Sign in Get Started

Docker ❤ WSL 2 – The Future of Docker Desktop for Windows

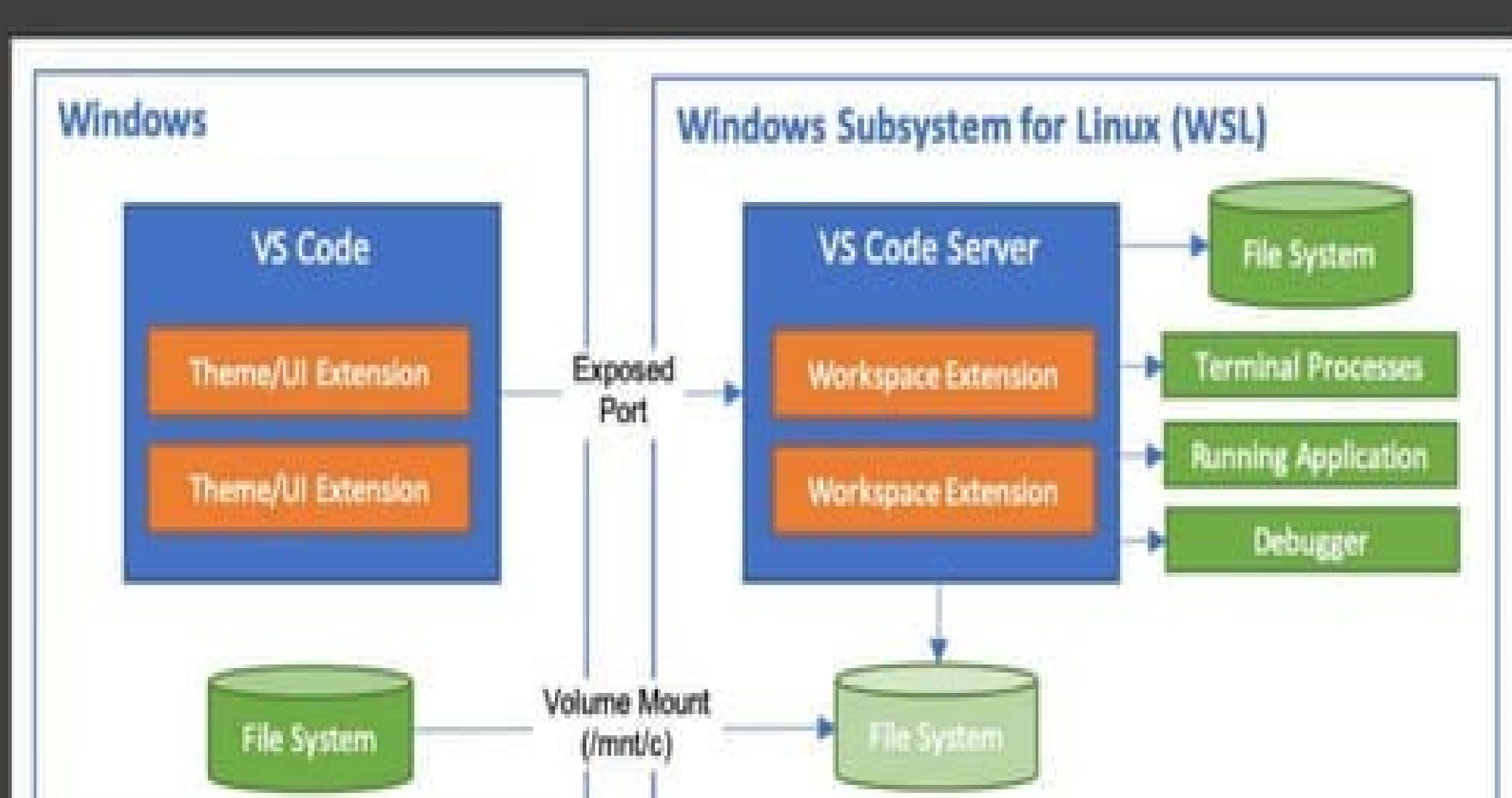
Desktop, Windows

SIMON FERQUEL
Jun 16

One of Docker's goals has always been to provide the best experience working with containers from a Desktop environment, with an experience as close to native as possible whether you are working on Windows, Mac or Linux. We spend a lot of time working with the software stacks provided by Microsoft and Apple to achieve this. As part of this work, we have been closely monitoring Windows Subsystem for Linux (WSL) since it was introduced in 2016, to see how we could leverage it for our products.

Visual Studio Code + WSL

- Visual Studio Code Remote
  - **Build**, **run**, and **debug** your Linux applications directly from VS Code

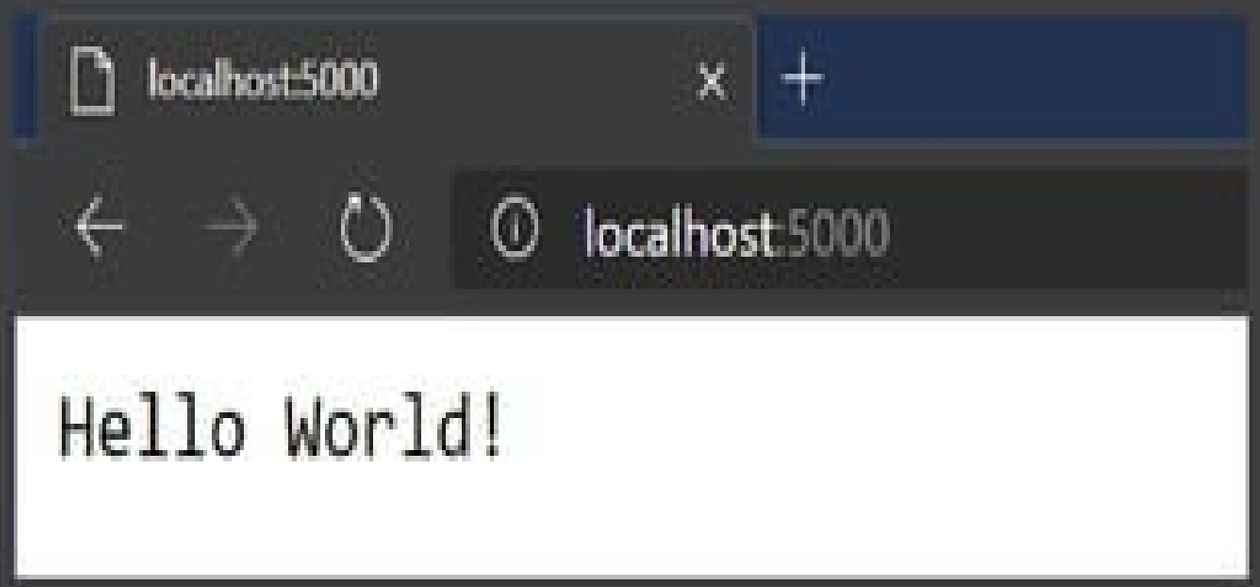


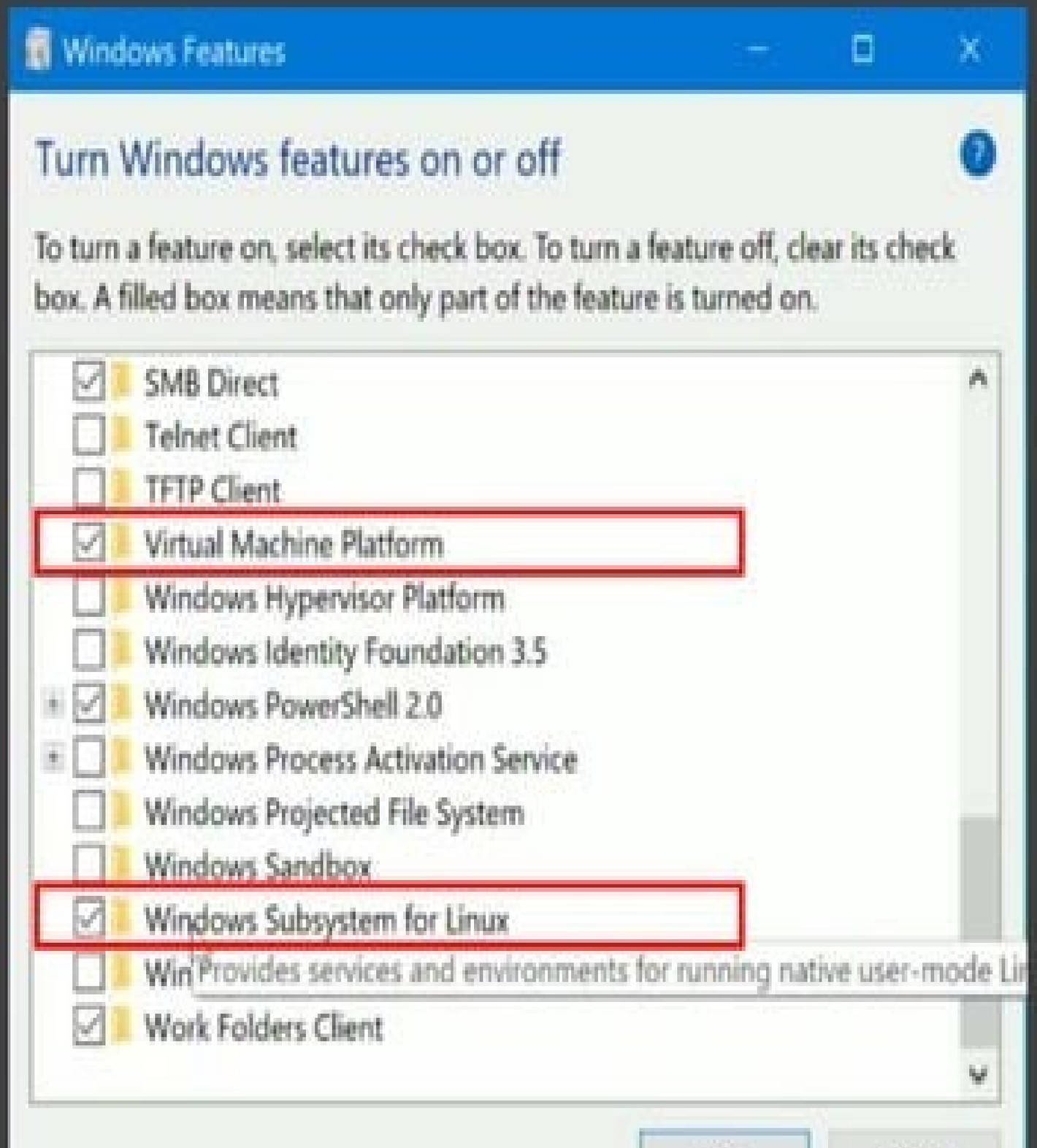
```
C:\sharpcode> echo "Hello" | clip.exe
C:\sharpcode> Get-Clipboard
Hello
```

```
$ echo "Hello from $(uname)" | clip.exe
```

```
C:\sharpcode> Get-Clipboard
Hello from Linux
```

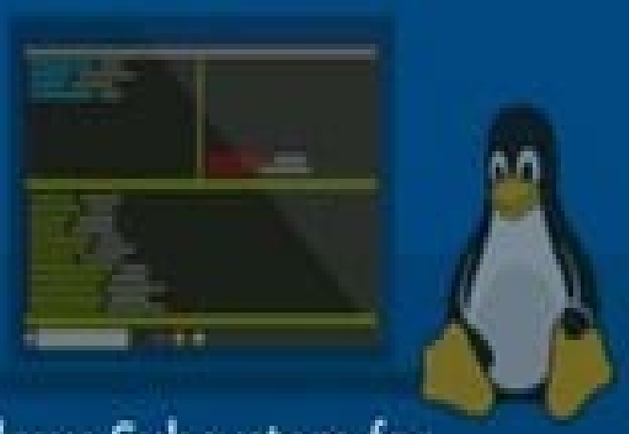
```
$ dotnet new mvc --name wsl-web
$ cd dotnet-web/
$ dotnet run
info: Microsoft.Hosting.Lifetime[0]
      Now listening on: https://localhost:5001
info: Microsoft.Hosting.Lifetime[0]
      Now listening on: http://localhost:5000
```




Windows Features
Turn Windows features on or off
To turn a feature on, select its check box. To turn a feature off, clear its check box. A filled box means that only part of the feature is turned on.
SMB Direct
Telnet Client
TFTP Client
Virtual Machine Platform
Windows Hypervisor Platform
Windows Identity Foundation 3.5
Windows PowerShell 2.0
Windows Process Activation Service
Windows Projected File System
Windows Sandbox
Windows Subsystem for Linux
Win
Provides services and environments for running native user-mode Li
Work Folders Client

Microsoft Store

Home Gaming Entertainment Productivity Deals Microsoft Search 1



Run Linux on Windows

Install and run Linux distributions side-by-side on the Windows Subsystem for Linux (WSL).

ubuntu® KALI BY OFFENSIVE SECURITY fedora remix

Docker under WSL 2

Agenda

- Containers are NOT VMs
- Working with Docker (Build, Ship, Run)
- But Why?
- Getting started
- Q & A

Containers are not VMs

Docker containers are NOT VMs

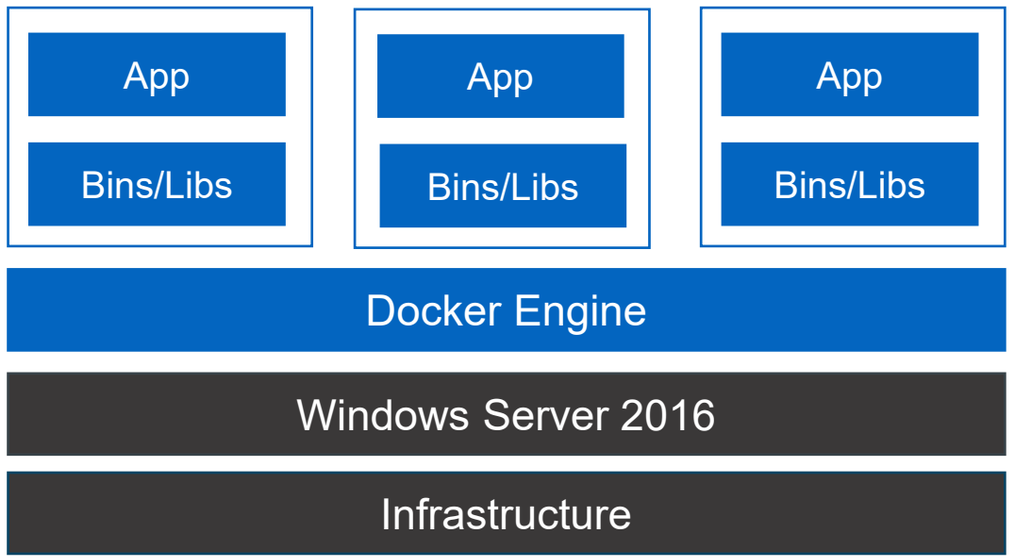
- Easy connection to make
- Fundamentally different architectures
- Fundamentally different benefits

VMs

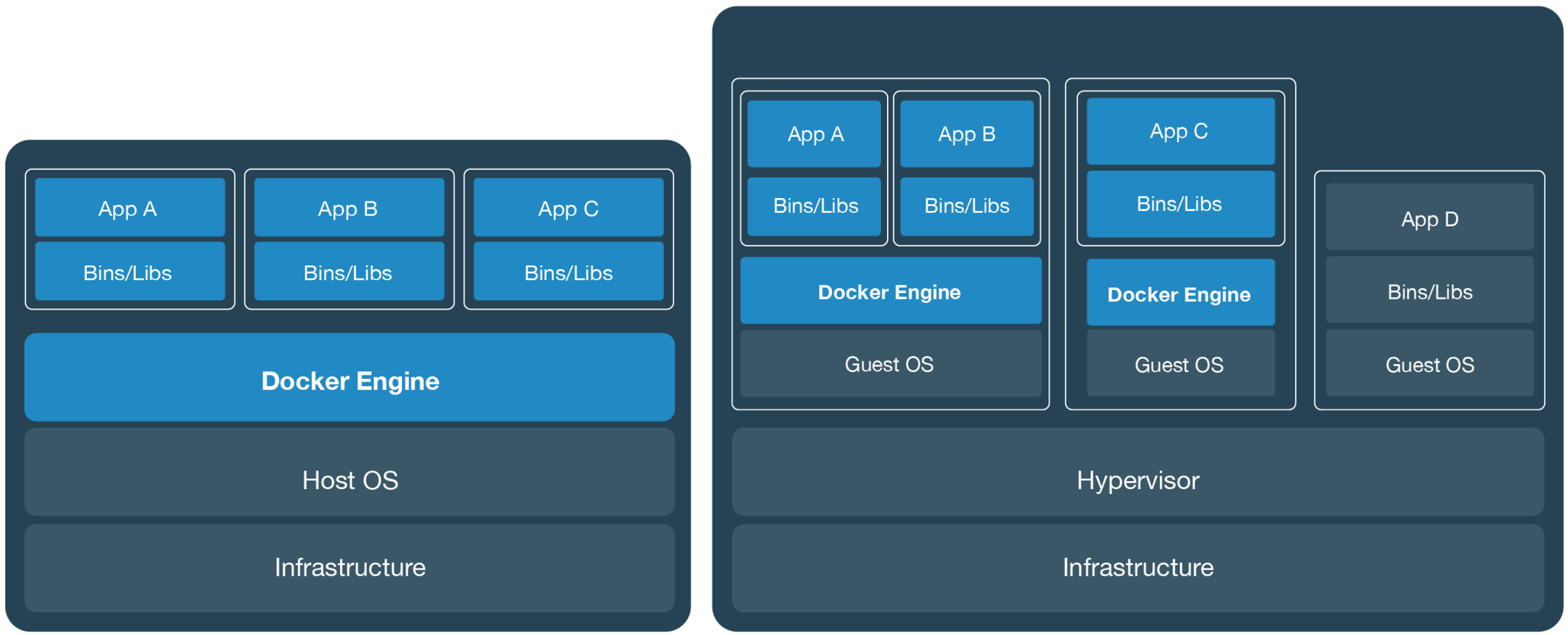
Containers

Docker + Windows Server = Windows Containers

- Native Windows containers powered by Docker Engine
- Windows kernel engineered with new primitives to support containers
- Deep integration with 2+ years of engineering collaboration in Docker Engine and Windows Server
- Microsoft is top 5 Docker open source project contributor and a Docker maintainer

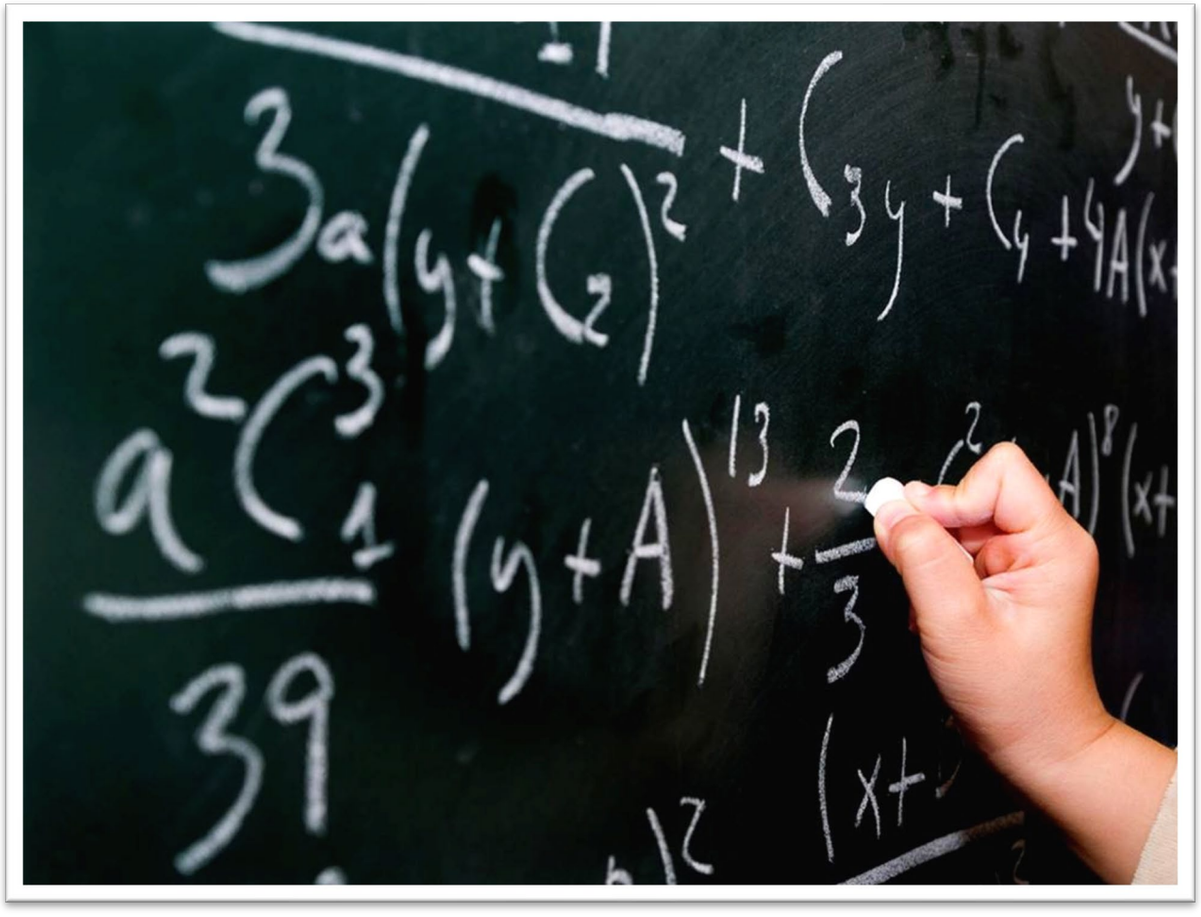


They’re different, not mutually exclusive



Variables to Consider

- Performance
- Security
- Scalability
- Existing Skillsets
- Costs
- Etc.



http://people-equation.com/do-your-words-encourage-or-deflate/math-equation_chalkboard/

Container Consolidation Testing

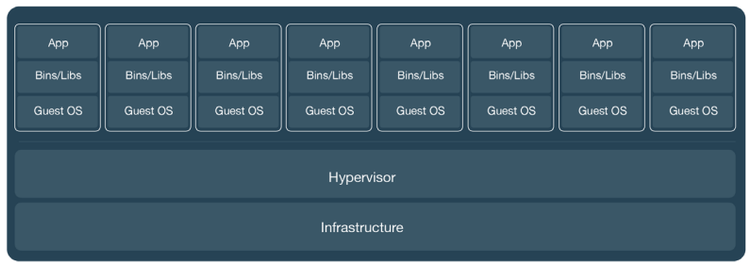
- **How can containers help organizations optimize hardware utilization?**

- Testing done by HPE, Docker and Industry Consultant
- Components:
  - Docker CS Engine 1.12.3
  - VMware ESXi 6
  - SysBench 1.0 (Nov 2016)
  - RHEL 7.2
  - HPE ProLiant DL360 Gen 9 servers with HPE 3PAR StorServ 8200 SSD Storage

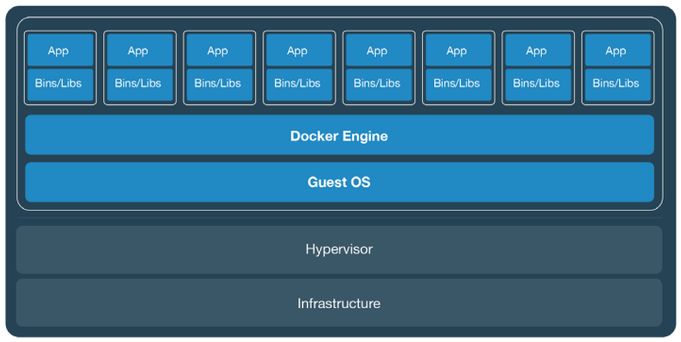
Testing Scenarios

- Measure SysBench performance across 3 configurations

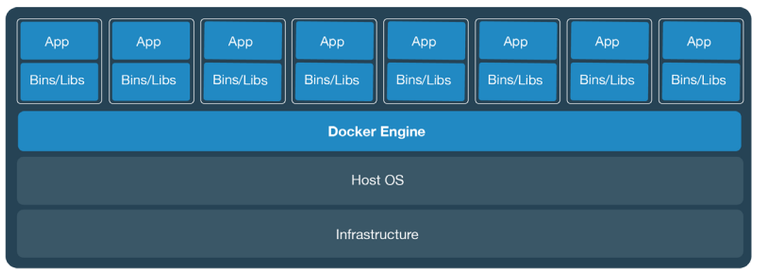
Need to replace this graphic w/ the one w/ 8 boxes



Virtual Machines



Virutal Machines with Containers



Containers on Bare Metal

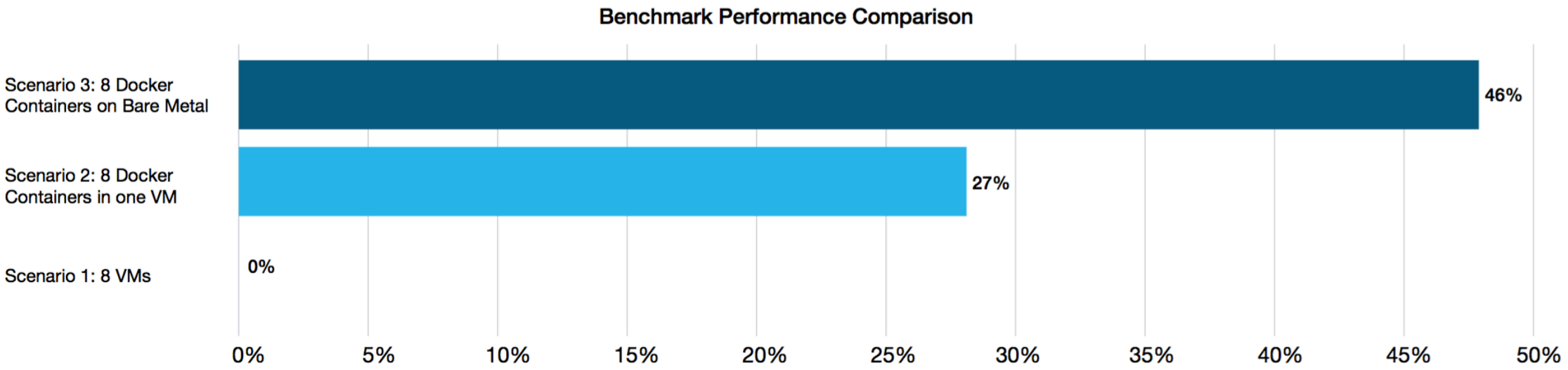
Scenario 1:
8 VMs

Scenario 2:
1 VM w/ 8 Containers

Scenario 3:
8 Containers on Bare Metal

Results

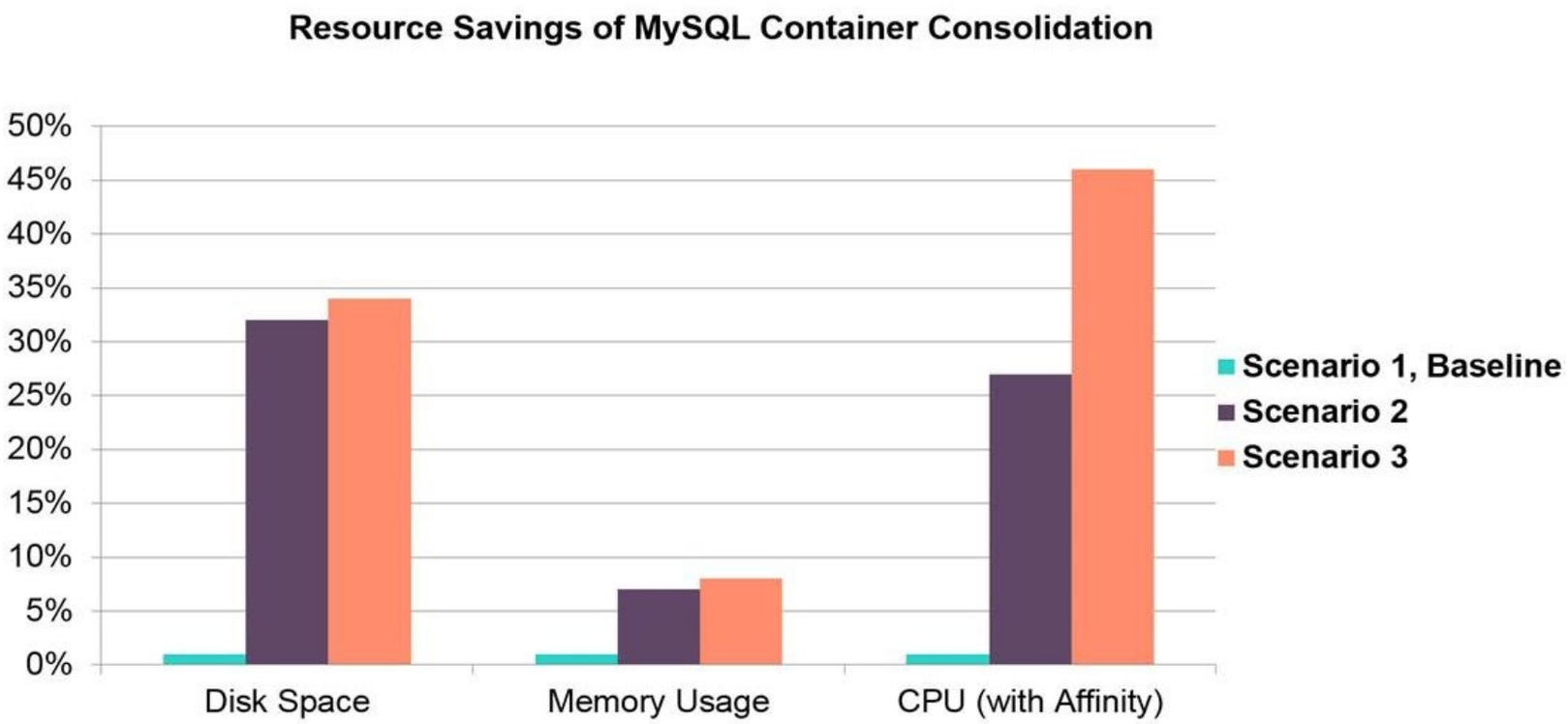
- Moving from VMs to containers increases performance 27% to 46%



Results are after VM and Container Tuning

Additional Savings

- Docker allowed for savings in memory and disk as well



Key Learnings

- Docker containers increase performance and flexiblity

1. Plan for Higher Density
2. Bare Metal or Bigger VMs
3. Tune To Optimize

Build, Ship, Run, Any App Anywhere

From Dev

To Ops

Any App

ASP.NET · .NET · Java · MORE

docker

Any OS

Windows · Linux

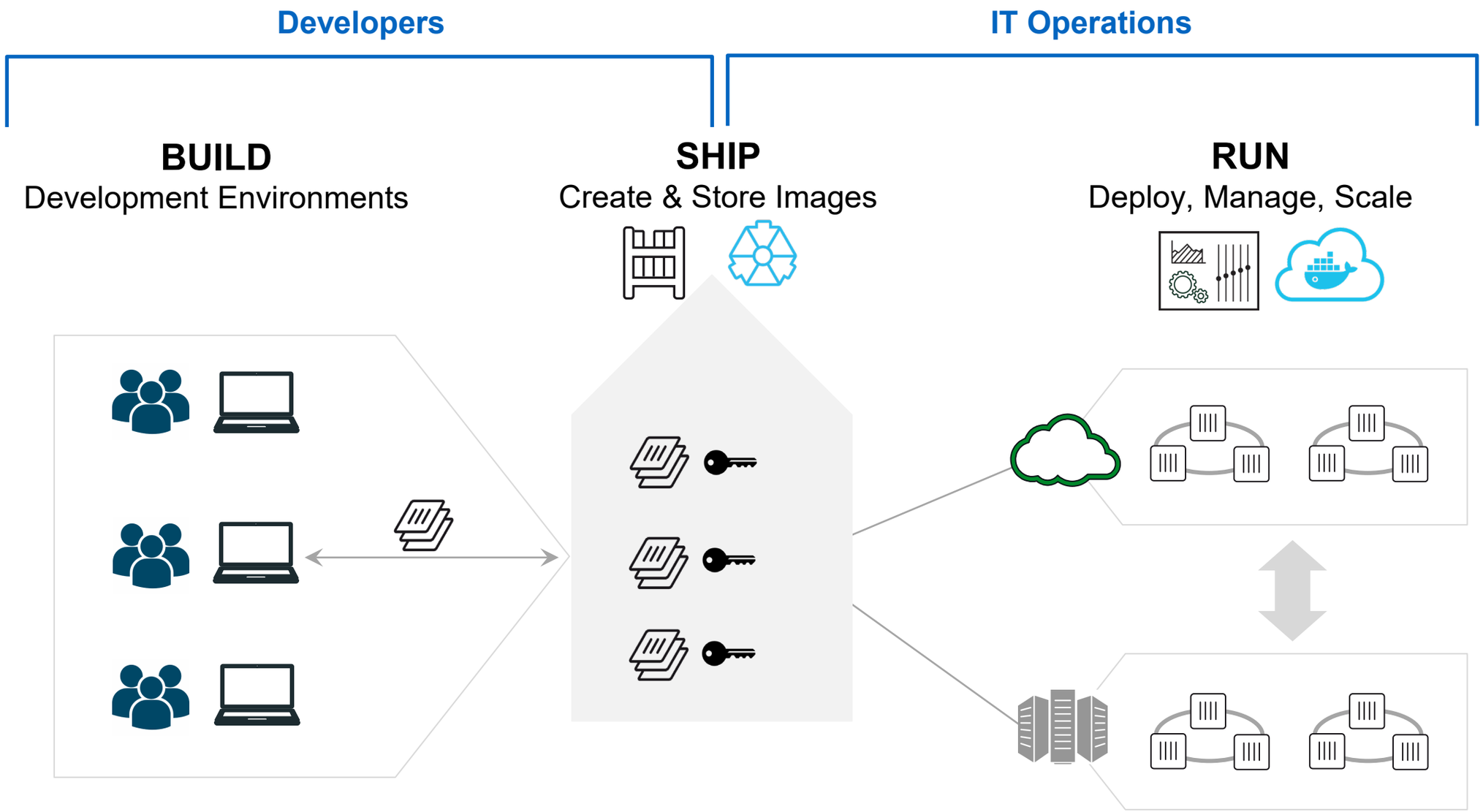
Anywhere

Physical · Virtual · Cloud

www.docker.com/enterprise

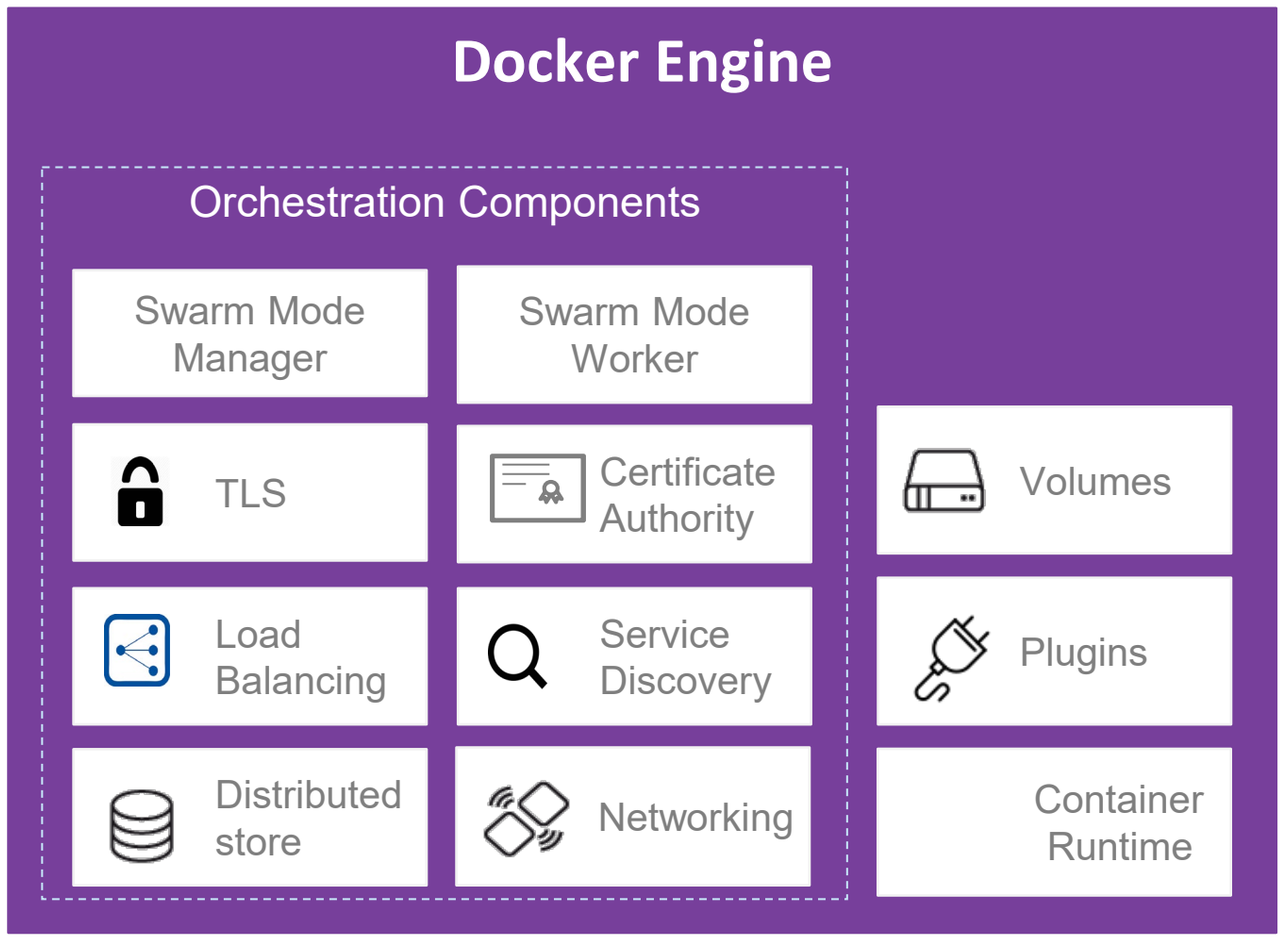
Build, Ship, and Run

Put it all together: Build, Ship, Run Workflow



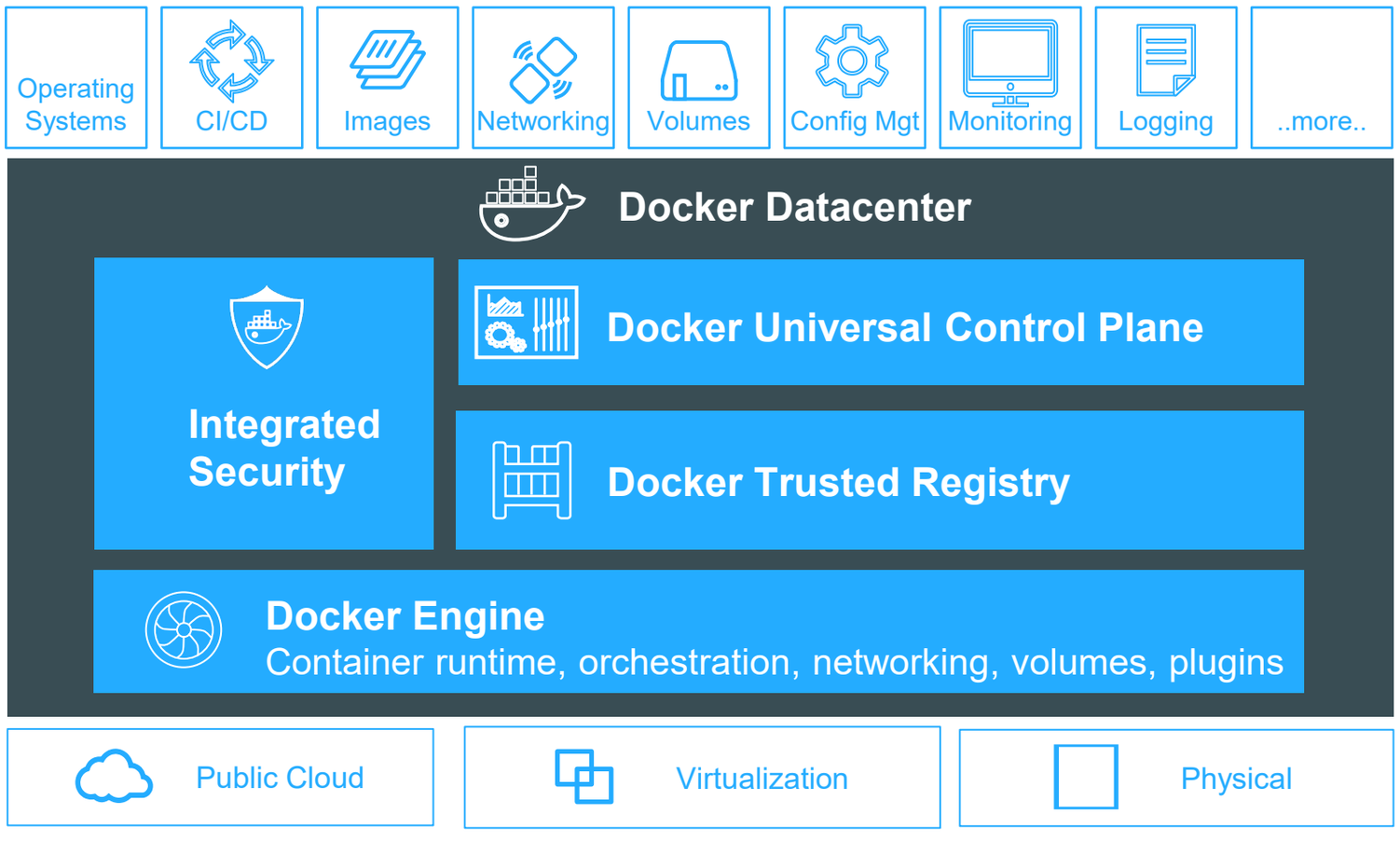
The building block: Docker Engine 1.12

Built in orchestration with scheduling, networking and scheduling

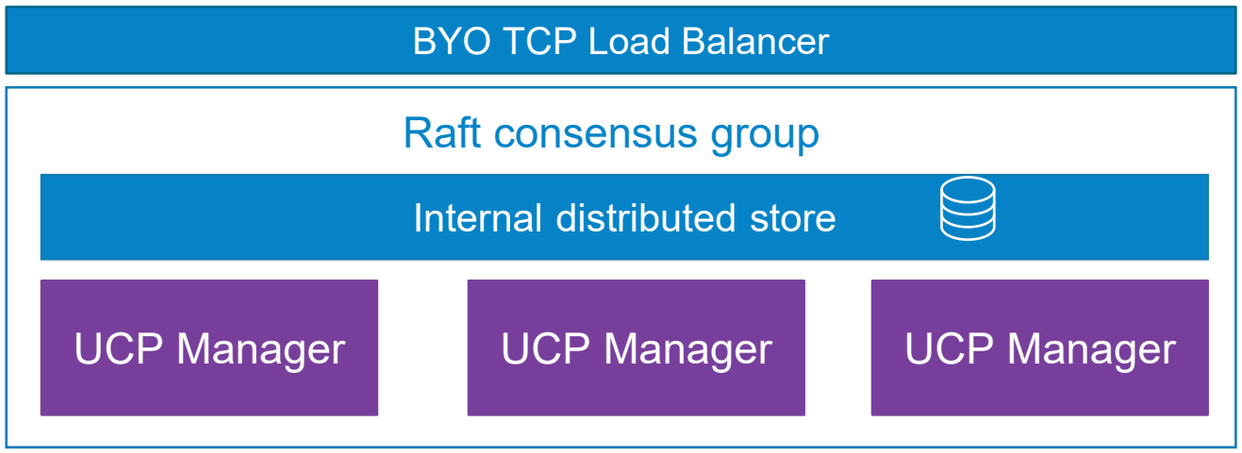


- Powerful yet simple, built in orchestration
- Declarative app services
- Built in container centric networking
- Built in default security
- Extensible with plugins, drivers and open APIs

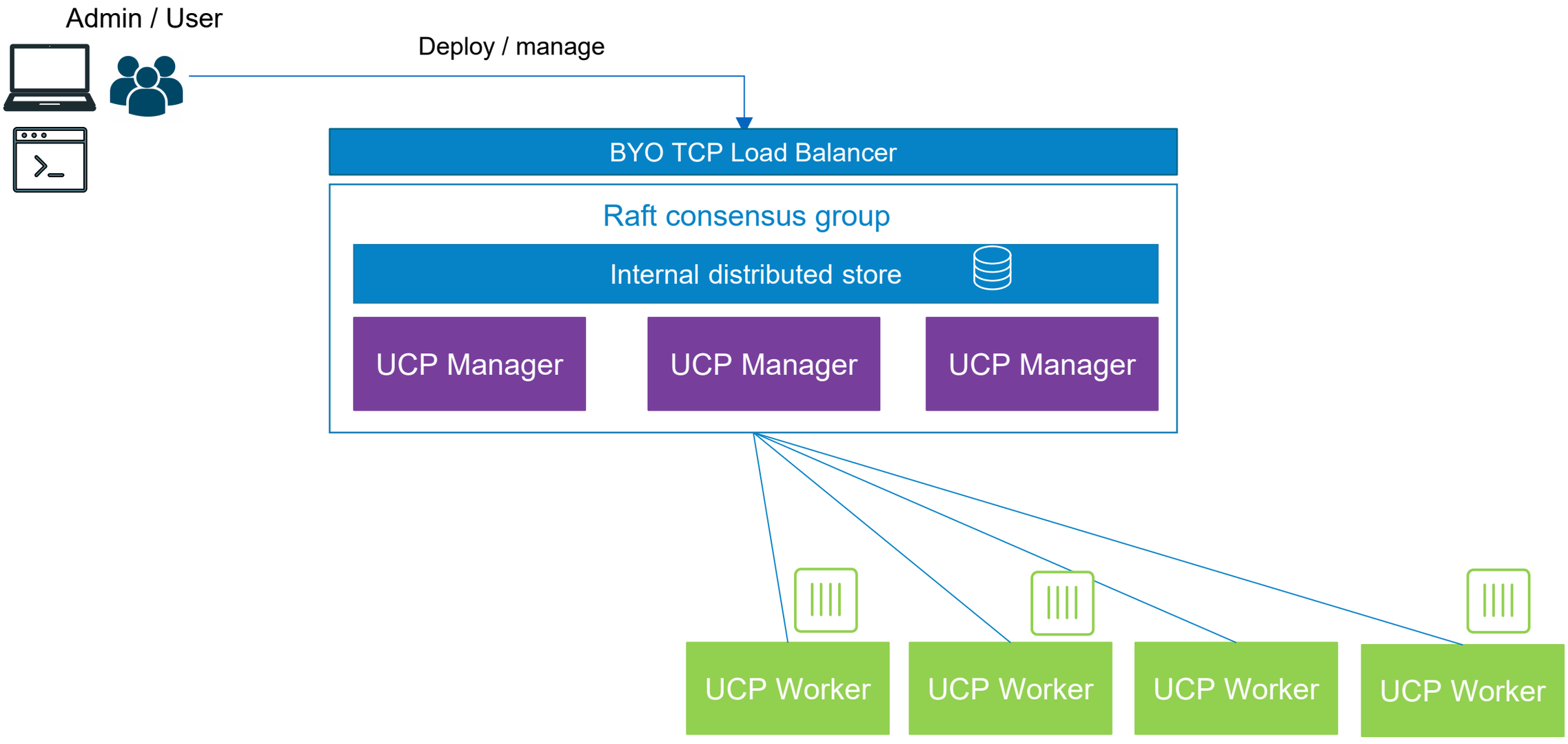
Docker Datacenter platform



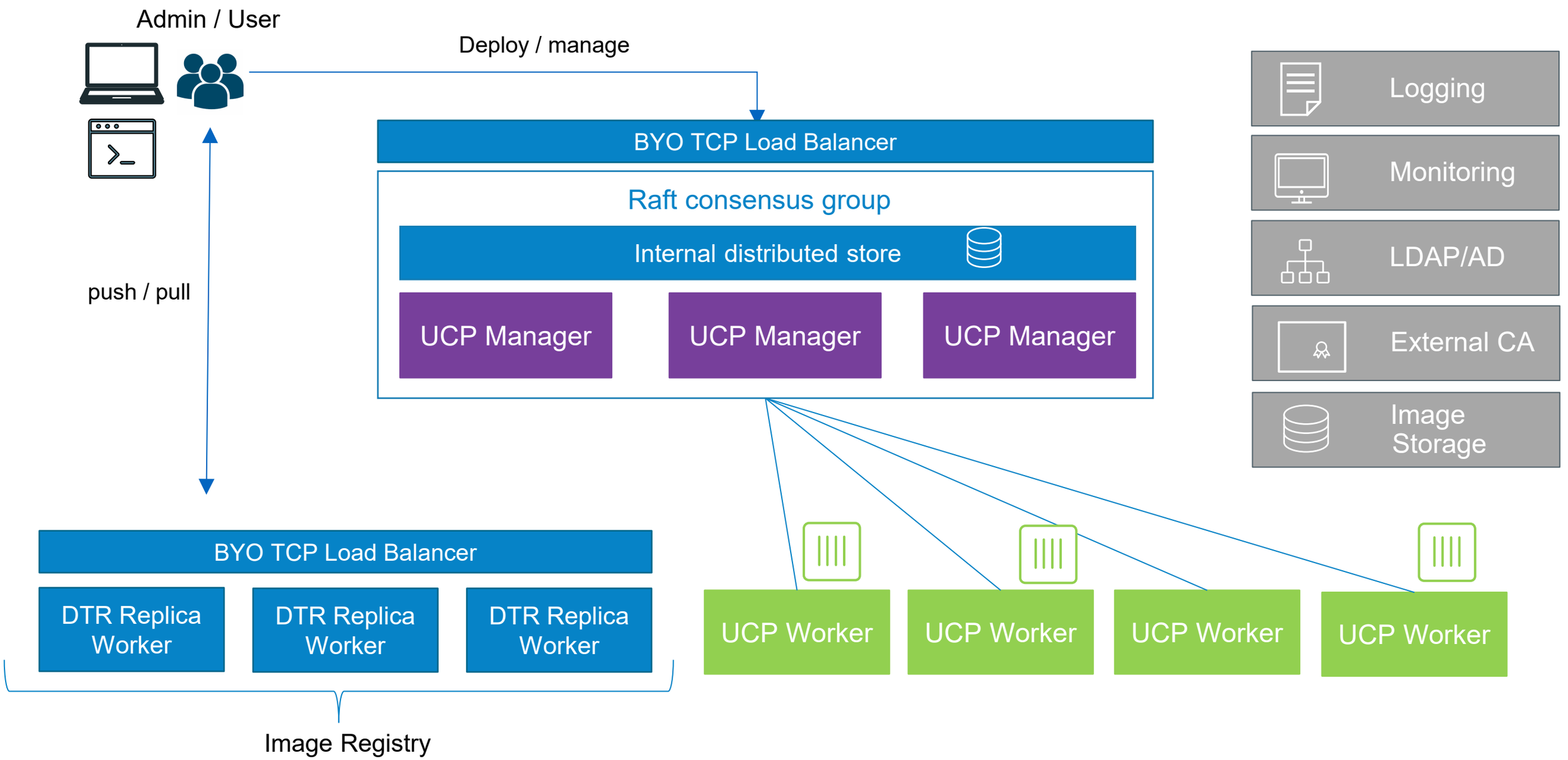
Docker Datacenter Architecture



Docker Datacenter Architecture



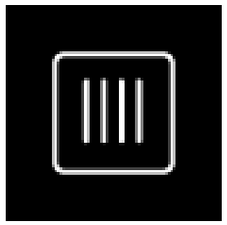
Docker Datacenter Architecture



Some Docker vocabulary

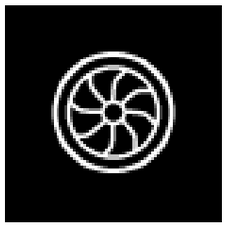
Docker Image

The basis of a Docker container. Represents a full application



Docker Container

The standard unit in which the application service resides and executes



Docker Engine

Creates, ships and runs Docker containers deployable on a physical or virtual, host locally, in a datacenter or cloud service provider

Registry Service (Docker Hub or Docker Trusted Registry)

Cloud or server based storage and distribution service for your images

Basic Docker Commands

```
$ docker pull mikegcoleman/catweb:1.0

$ docker images

$ docker run –d –p 5000:5000 –-name catweb mikegcoleman/catweb:latest

$ docker ps

$ docker stop catweb (or <container id>)

$ docker rm catweb (or <container id>)

$ docker rmi mikegcoleman/catweb:latest (or <image id>)

$ docker build –t mikegcoleman/catweb:2.0 .

$ docker push mikegcoleman/catweb:2.0
```

Dockerfile – Linux Example

```
# our base image
FROM alpine:latest

# Install python and pip
RUN apk add --update py-pip

# upgrade pip
RUN pip install --upgrade pip

# install Python modules needed by the Python app
COPY requirements.txt /usr/src/app/
RUN pip install --no-cache-dir -r /usr/src/app/requirements.txt

# copy files required for the app to run
COPY app.py /usr/src/app/
COPY templates/index.html /usr/src/app/templates/

# tell the port number the container should expose
EXPOSE 5000

# run the application
CMD ["python", "/usr/src/app/app.py"]
```

- Instructions on how to build a Docker image
- Looks very similar to “native” commands
- Important to optimize your Dockerfile

Dockerfile – Windows Example

19 lines (15 sloc) | 832 Bytes

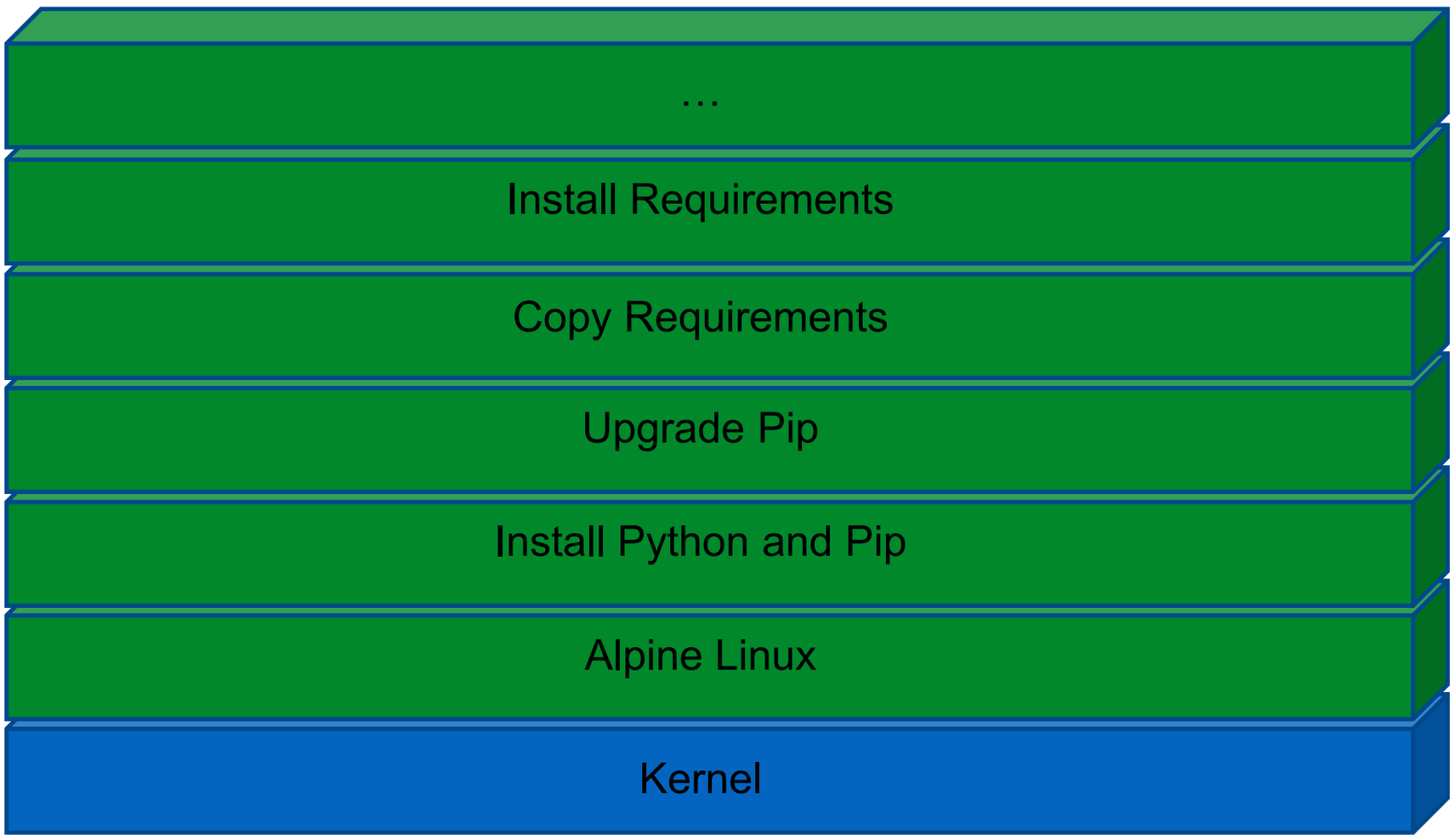
Raw | Blame | History

```
FROM microsoft/windowsservercore

ENV NPM_CONFIG_LOGLEVEL info
ENV NODE_VERSION 6.5.0
ENV NODE_SHA256 0c0962800916c7104ce6643302b2592172183d76e34997823be3978b5ee34cf2

RUN powershell -Command \
    $ErrorActionPreference = 'Stop' ; \
    (New-Object System.Net.WebClient).DownloadFile('https://nodejs.org/dist/v%NODE_VERSION%/node-v%NODE_VERSION%-win-x64.zip',
    if ((Get-FileHash node.zip -Algorithm sha256).Hash -ne $env:NODE_SHA256) {exit 1} ; \
    Expand-Archive node.zip -DestinationPath C:\ ; \
    Rename-Item 'C:\node-v%NODE_VERSION%-win-x64' 'C:\nodejs' ; \
    New-Item '%APPDATA%\npm' ; \
    $env:PATH = 'C:\nodejs;%APPDATA%\npm;' + $env:PATH ; \
    [Environment]::SetEnvironmentVariable('PATH', $env:PATH, [EnvironmentVariableTarget]::Machine) ; \
    Remove-Item -Path node.zip

CMD [ "node.exe" ]
```

Demo

Build, Ship, and Run

Image Layers



Docker File System

- Logical file system by grouping different file system primitives into branches (directories, file systems, subvolumes, snapshots)
- Each branch represents a layer in a Docker image
- Allows images to be constructed / deconstructed as needed vs. a huge monolithic image (ala traditional virtual machines)
- When a container is started a writeable layer is added to the “top” of the file system

Copy on Write

- Super efficient:
  - Sub second instantiation times for containers
  - New container can take <1 Mb of space

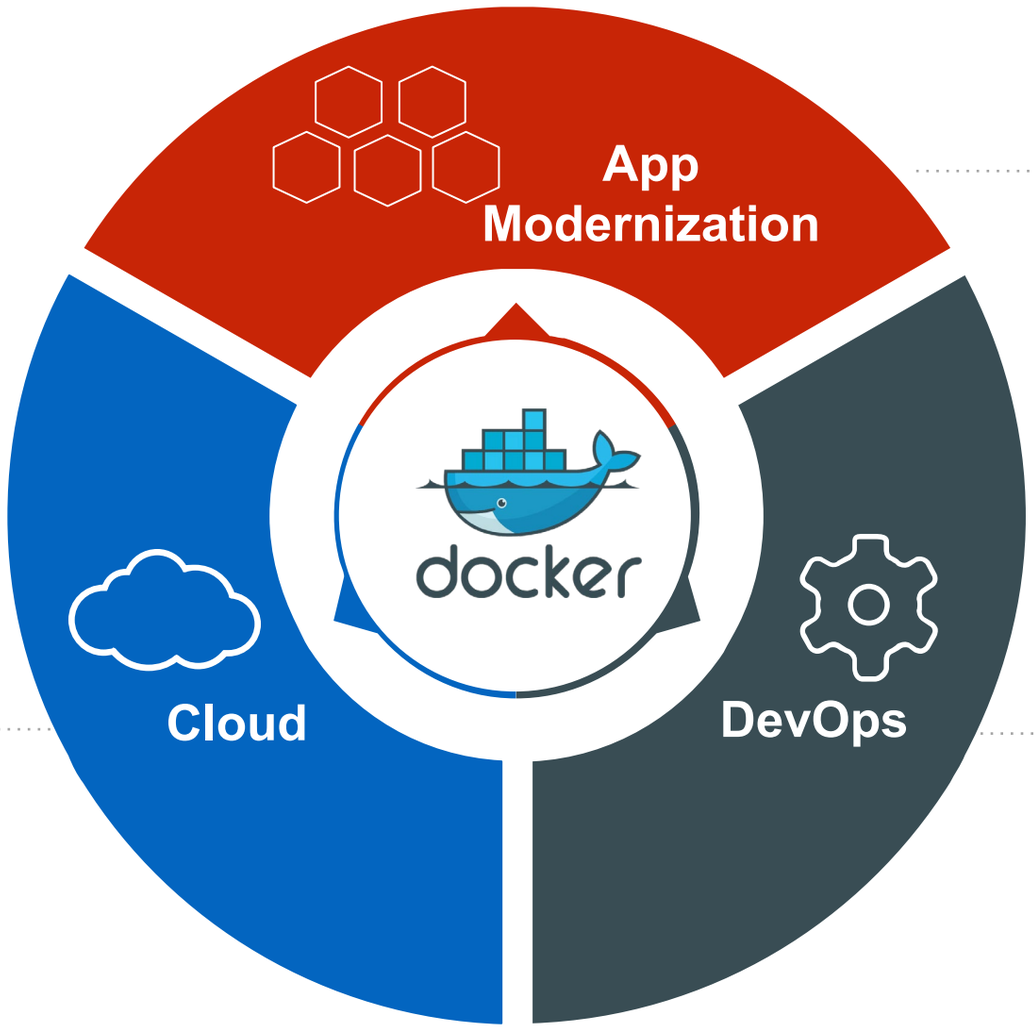
- Containers appears to be a copy of the original image
- But, it is really just a link to the original shared image
- If someone writes a change to the file system, a copy of the affected file/directory is “copied up”

What about data persistence?

- Volumes allow you to specify a directory in the container that exists outside of the docker file system structure
- Can be used to share (and persist) data between containers
- Directory persists after the container is deleted
  - Unless you explicitly delete it
- Can be created in a Dockerfile or via CLI

But Why?

Enterprises are looking to Docker for critical transformations



3 out 4

Top initiatives revolve around applications

80%

Docker is central to cloud strategy

44%

Looking to adopt DevOps

State of App development Survey: Q1 2016

Docker delivers speed, flexibility and savings

Agility

13X

More software releases

65%

Reduction in developer onboarding time

Portability

41%

Move workloads across private/public clouds

Eliminate

“works on my machine” issues

Control

62%

Report reduction in MTTR

10X

Cost reduction in maintaining existing applications

State of App development Survey: Q1 2016, Cornell University case study

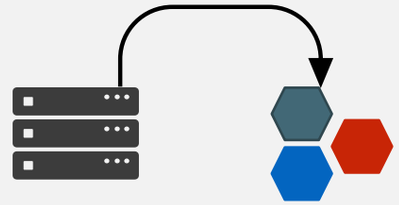
One platform delivers one journey for all applications

1

Containerize Legacy Applications
Lift and shift for portability and efficiency

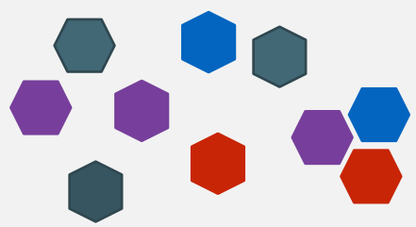
2

Transform Legacy to Microservices
Look for shared services to transform

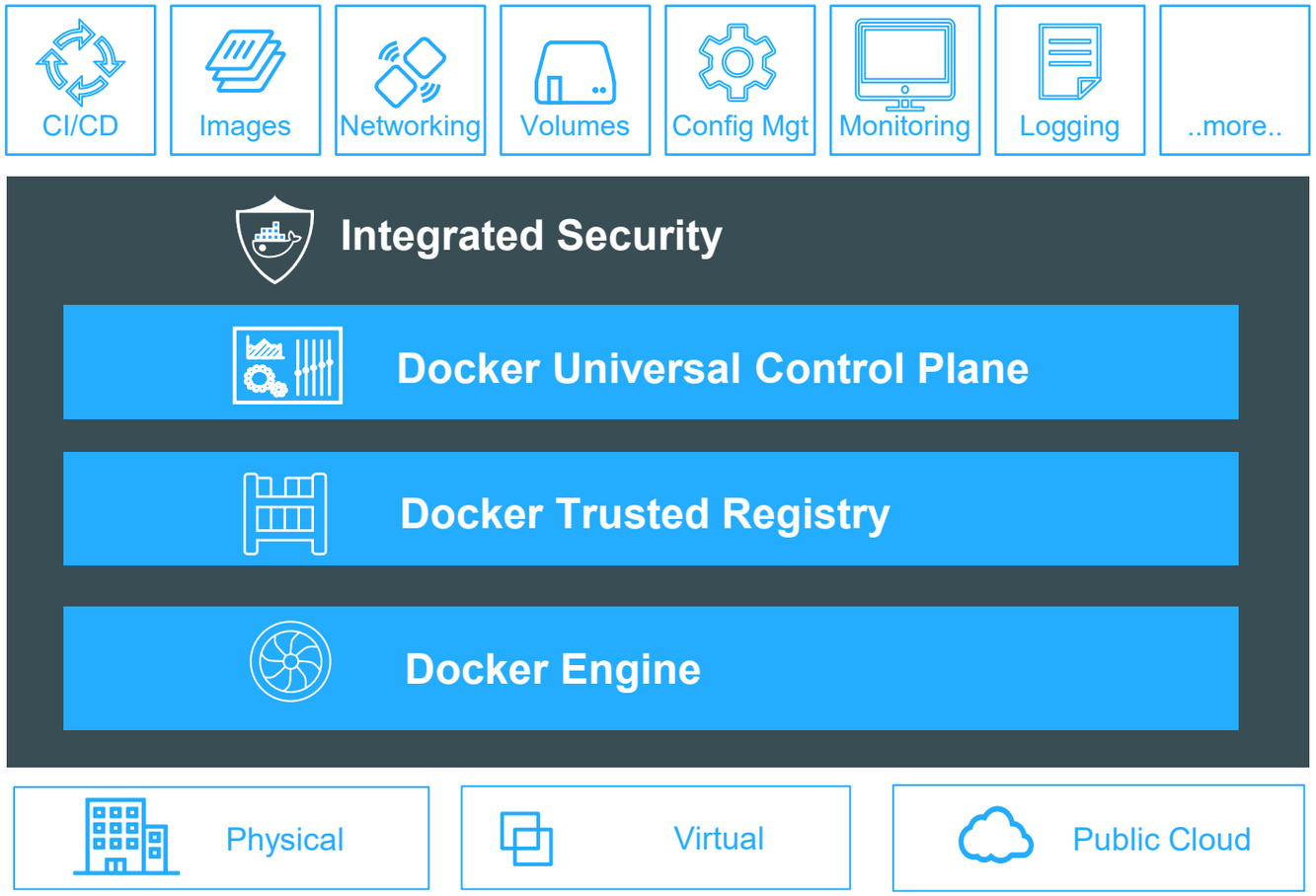


3

Accelerate New Applications
Greenfield innovation



Containers in production with Docker Datacenter



- Enterprise container orchestration, management and security for dev and ops
- Available today for Linux environments
- Q4 2016 beta for Windows environments

Getting Started

Docker on Linux

- Create a Linux VM (or use physical), and install Docker
  - Requires kernel 3.10

- Stable builds
  - `curl -sSL https://get.docker.com/ | sh`

- Test and experimental builds
  - `curl -sSL https://test.docker.com/ | sh`
  - `curl -sSL https://experimental.docker.com/ | sh`

- Can also manually install (see docs)

Docker for Windows / Mac

- Currently in public beta
- Easy to install: Get up and running on Docker in minutes
- Leverages Hyper-V (Windows) or xhyv (Mac)
  - Docker for Windows requires Windows Pro 10, Enterprise, or Education
- Full API / CLI compatibility
- OS integration for increased stability and speed

Docker for Azure / AWS

- Currently in public beta
  - https://beta.docker.com/

- Easily deploy Docker 1.12 Swarm clusters (Linux)

- Scale up and down easily

- Integrate with underlying platform (i.e. load balancers)

Walk, Jog, Run

Walk:

- Setup your preferred Docker environment
- Fire up some prebuilt images (nginx, hello-world, mikegcoleman/catweb)

Jog:

- Pick a well documented solution (Wordpress, Jenkins, etc)
- Build it for yourself (blogs are your friend)

Run:

- Extend one your Walk solution or Dockerize an existing project
- Build your own Dockerfiles
- Experiment with Docker Compose and Swarm Mode

Hands-on Labs

http://github.com/docker/labs

Dockers Guide

Docker Guides

https://xuanthulab.net/gioi-thieu-va-cai-dat-wsl-tren-windows.html

https://github.com/vietnam-devs/coolstore-microservices/blob/master/docker-compose.yml

https://github.com/dockerframework/phpmyadmin/blob/master/docker-compose.yml

https://gist.github.com/dracken/de0c495572f600b8b2f0190306946332

https://github.com/osixia/docker-phpMyAdmin

CÁM ƠN ĐÃ LẮNG NGHE!